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A Crowdsensing-Based Framework for Indirect Bridge Monitoring Using Mel-Frequency Cepstral Analysis Considering Elimination of Operational Effects

Nima Shirzad-Ghaleroudkhani, Aff.M.ASCE[1]; and Mustafa Gül, A.M.ASCE[2]

Abstract: This paper puts forward an indirect bridge monitoring method using Mel-frequency cepstral analysis of inverse-filtered drive-by acceleration signals collected through smartphones. Crowdsensing-based approaches using data collected by smart cars and smartphones opened a new chapter in bridge monitoring by reducing the costs and increasing the efficiency of the bridge monitoring process. However, the major challenge of the dominancy of the operational effects in the recorded drive-by vibrations overshadows the bridge monitoring objective. This paper proposes an inverse filtering-based monitoring approach to suppress operational effects. The inverse-filtered spectrum is later employed in a Mel-frequency cepstral analysis, leading to the calculation of the abnormality index, which is then used to detect the change in the bridge state. The performance of the proposed method in suppressing operational effects is assessed through a series of laboratory and real-life experiments. Afterward, the damage detection capability of the method is investigated for two damage levels at different locations along the bridge, modeled in a laboratory environment. The results provide evidence for the capability of the proposed method in drive-by damage detection of bridges. Moreover, using the smartphone as the data acquisition device paves the path toward the implementation of the method for crowdsensing-based bridge monitoring in future smart cities, although more operational factors such as passenger interactions and resulting smartphone motions need to be considered in future studies. **DOI: 10.1061/JSENDH.STENG-11748.** *© 2023 American Society of Civil Engineers.*

Author keywords: Indirect bridge monitoring; Drive-by acceleration signal; Eliminating operational effects; Inverse filtering method; Mel-frequency Cepstral analysis; Smartphone.

Introduction

Bridges are critical elements of transportation infrastructure, yet as exposed structures, they are prone to continuous degradation due to environmental effects (Andrade et al. 2002; Guettala and Abibsi 2006; Liu and Weyers 1998). Climate change has hastened the deterioration of bridge structures in recent decades (Xie et al. 2011), necessitating more frequent inspection and maintenance. According to a recent assessment by the ASCE Infrastructure Report Card (ASCE 2021), about 7.5% of bridges in the United States are structurally deficient, with 42% older than 50 years. A similar report by CIRC (2019) showed that nearly 40% of roads and bridges in Canada are in fair, poor, or very poor conditions. These reports attest to the fact that the resilience of the transportation system will be undermined without a proper bridge monitoring network.

Traditional bridge monitoring methods relied on field tests through visual inspections. Later, with the development in technology, accurate devices were available to engineers to measure bridge response, including accelerometers, strain gauges, and radar sensors. Structural health monitoring (SHM) approaches were then developed focusing on employing such devices to detect structural anomalies (Hsieh et al. 2006; Nazarian et al. 2016; Santini Bell et al. 2013; Wenzel 2009). Although these methods were proved to be effective, direct data collection from bridge structures could be challenging. The complexity of the installation of the devices as well as the necessity for power and data storage networks are major concerns when dealing with a population of bridges. Moreover, during the course of the project, the serviceability of the bridges is frequently constrained by lane closures (Ramadan and Sisiopiku 2016), which may negatively impact the capacity of the transportation system.

Researchers have recently become interested in an alternate monitoring strategy that uses an instrumented vehicle to collect vibrational data. Initially studied by Yang et al. (Lin and Yang 2005; Yang et al. 2004), drive-by approaches offer low-cost and efficient opportunities for indirectly monitoring bridges without the need for direct equipment installation, albeit introducing new challenges compared to direct methods. Mixed vibrations recorded on the vehicle are highly dominated by vehicle vibrations, including engine, suspension system, tire effects, and the effect of road roughness. Extracting bridge dynamic features from those mixed vibrations was the main concern of indirect bridge monitoring studies in recent years (Elhattab 2017; Malekjafarian et al. 2015; Sitton et al. 2020b; Yang and Yang 2018). These studies are thoroughly reviewed in the next section.

In addition to the capability of monitoring multiple bridges using a single instrumented vehicle, drive-by approaches pave the path for the implementation of crowdsensing-based methods for bridge health monitoring. Nowadays, smartphones contain powerful sensors, including accelerometers and gyroscopes, that could be

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used to collect vibrational data from vehicles traveling over bridges. Hence, without the need to use dedicated instrumented vehicles, a large amount of crowdsensed data are available to monitor the real-time state of all the bridges in a metropolitan area. In this regard, many studies (Matarazzo et al. 2018; Mei et al. 2020; Shirzad-Ghaleroudkhani et al. 2020, 2022; Sitton et al. 2020a) focused on the application of smartphones for crowdsensing-based bridge monitoring.

This study concentrates on the detection of damage on the bridge using drive-by collected acceleration signals from passengers' smartphones considering operational effects. Note that the operational factors investigated in this paper are related to vehicle and road properties, while passenger interactions and the resulting smartphone motions are not in the scope of this study. The proposed approach is built upon two main methods: Inverse filtering and Mel-frequency cepstral analysis. In the following, first, a literature review on indirect bridge monitoring is provided. Then, the methodology is explained in detail. Afterward, the performance of the method in eliminating operational effects is investigated through laboratory experiments and also under real-life conditions. Then, the damage detection capability of the proposed method is assessed in laboratory conditions.

Background

This section reviews previous studies focusing on monitoring bridges using mixed vibration recorded on vehicles. These studies are classified into bridge frequency detection, mode shape detection, force- and displacement-based damage detections, and cepstral analysis.

In addition, major challenges in those studies and how the scope of this study aligns with those limitations and gaps are explained.

Extracting bridge frequencies was the focus of a plethora of studies, including the groundbreaking study by Yang et al. (2004). These studies aimed at addressing the challenge of detecting bridge frequencies in the spectrum of mixed drive-by recorded vibrations. Lin and Yang (2005) investigated the potential of using a tractor-trailer system to detect the fundamental frequency of a bridge in real-life conditions. They concluded that lower speeds of the vehicle provide higher resolution spectra, which are more suitable for frequency detection, while reducing the excitation level leads to a reduced bridge presence in the recorded mixed vibrations. On the other hand, higher speeds result in the dominance of road roughness in the vibrations, which too fades bridge presence in the spectrum. Keenahan et al. (2013) provided a solution for that issue by proposing the subtraction of the vibrations recorded on the two axles of the vehicle. They showed that the road roughness effect could be substantially reduced using their proposed method. Later, other studies employed novel approaches for detecting bridge frequencies, including empirical mode decomposition (EMD) (Yang and Chang 2009), generalized pattern search algorithm (GPSA) (Li et al. 2014), and frequency domain decomposition (FDD) (Malekjafarian and O'Brien 2014a). All these studies arrive at the same conclusion that the presence of the bridge in drive-by vibrations is not significant and that the effect of vehicle and road roughness factors needs to be considered.

Another group of studies took the next step toward detecting bridge mode shapes. Mode shapes could be decisive in bridge monitoring as any discontinuity in the mode shape could reveal both the intensity and location of the damage (Arora et al. 2009; Huth et al. 2005; Pandey et al. 1991). Among the first studies in this field, Zhang et al. (2012) employed a shaker-equipped vehicle model to approximate the mode shape of the bridge. Although the level of accuracy of their method was obtained for low vehicle speeds of less than 18 km/hr, it was a remarkable study that showed the potential of extracting bridge mode shapes using indirect methods. Afterward, researchers employed different methodologies to increase the accuracy of indirect bridge mode shape detection, using Hilbert transform (HT) (Yang et al. 2014), singular value decomposition (SVD) (Oshima et al. 2014), short-time frequency domain decomposition (STFDD) (Malekjafarian and OBrien 2014b), and matrix completion methods (Mei et al. 2021). Most of these studies conducted low-speed vehicle passages over the bridge to reduce the effect of noise and operational effects.

Detection of damage on the bridge is not limited to the identification of modal parameters. For instance, O'Brien et al. (2014) utilized moving force identification (MFI), a method based on finding the time history of forces applied to the bridge, to detect damage on the bridge. Zhang et al. (2013) employed the operating deflection shape curvature (ODSC) method. Yin and Tang (2011) used proper orthogonal decomposition (POD) to calculate the difference between the displacement responses of a vehicle passing the damaged and the healthy bridges. In addition, multiple studies (Khorram et al. 2012; McGetrick and Kim 2013, 2014) focused on the application of wavelet transform, and machine-learning methods were employed in some other studies (Lederman et al. 2014; Liu et al. 2019) for bridge damage detection. Recently, a novel damage detection approach based on Mel-frequency cepstral analysis has been proposed in the authors' research group (Mei et al. 2019; Mei and Gül 2019).

Cepstrum, first introduced by Childers et al. (1977), is defined as the inverse Fourier transform of the logarithm of the spectrum of a signal. Initially developed to study seismic echoes from earthquakes and explosions, it became a well-known audio processing (Alexandre and Lockwood 1993; Furui 1981; Tokuda et al. 1995) tool. Mel-frequency cepstral (MFC) analysis is among the most popular cepstral-based signal processing methods, and efficiency in speech recognition has been widely investigated (On et al. 2006; Vergin et al. 1999; Zheng et al. 2001). Instead of focusing on peak analysis, such as natural frequency detection, MFC analysis extracts patterns in the spectrum of a signal through extracting MFC features, called Mel-frequency cepstral coefficients (MFCCs). In other words, MFCCs are a vector of features representing patterns in the spectrum of a signal. MFC has been employed in SHM studies in recent years. For instance, Zhang et al. (2011) proposed a delamination detection method using MFFCs to remove noise and subjectivity. Moreover, Balsamo et al. (2014) employed MFCCs as damage features and showed that the accuracy in damage detection is higher using MFCCs compared to using autoregressive coefficients. Regarding indirect bridge monitoring, Mei et al. (2019) used extracted MFCCs and detected damage on the bridge by applying principal component analysis (PCA), which provided insights into both the presence and severity of the damage.

The main challenge facing indirect bridge monitoring, as mentioned before, is the presence of operational effects, primarily vehicle and road properties. All previously mentioned studies focused on capturing bridge properties in drive-by vibrational signals, which are dominated by operational effects, through a variety of methods. One common element of all those studies was focusing on the on-bridge vibrations alone, i.e., when the vehicle is moving on the bridge. Looking from a different point of view, the authors (Shirzad-Ghaleroudkhani and Gül 2020) have recently proposed employing the off-bridge signals, i.e., signals recorded when the vehicle is traveling off the bridge and on the ground, together with the on-bridge signals. Off-bridge signals contain crucial information about the operational factors, especially the vehicle, which could be employed to design a filter with the focus of eliminating

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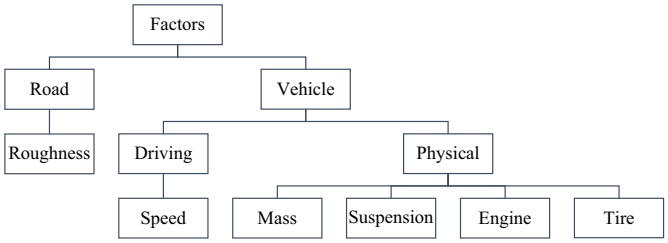


Fig. 1. Variety of factors affecting drive-by vibrations.

operational effects from the on-bridge signals. Inverse filtering proposed in that study (Shirzad-Ghaleroudkhani and Gül 2020) was successfully used for bridge frequency detection in a controlled laboratory environment, which was later enhanced and applied in real-life conditions for two full-scale bridges (Shirzad-Ghaleroudkhani and Gül 2021). The data acquisition device in those studies was a smartphone, which aligned with the perspective toward future applications for smart crowdsensing-based bridge monitoring.

The scope of this paper is to (1) study the significance of operational factors on the results from MFC analysis, and (2) investigate the performance of combining the inverse filtering method with MFC analysis in addressing this challenge. Data acquisition devices in this study will only be smartphones, while potential passenger interactions and resulting smartphone motions are not among the operational effects considered in this study and need to be investigated in the future. In the proposed approach in this paper, first, inverse filtering is first applied to suppress operational effects in drive-by vibrations. Later, an abnormality index is defined based on the distribution of MFCCs compared to a baseline healthy case, representing the change in the bridge dynamic state.

Methodology

In this section, two analytical aspects of the proposed methodology are outlined. First, the inverse filtering approach is described, which focuses on the elimination of operational effects from collected drive-by vibrations. Afterward, the MFC analysis is discussed, followed by introducing an abnormality index to account for the change in the bridge state.

Inverse Filtering Methodology

Focusing on the elimination of operational effects from drive-by collected vibrations, the inverse filtering method utilizes the off-bridge vibrations as a tool to design the filter. Drive-by vibration is affected by a variety of factors, as presented in Fig. 1. Since the bridge only affects the recorded vibrations during the on-bridge portion of the vehicle's travel, whereas other factors are present at all times, it would be possible to create an appropriate filter to amplify bridge effect while suppressing the rest of the factors. Such a filter could be designed using off-bridge data.

In terms of frequency domain analysis, an inverse filter is designed by taking the mathematical inverse of the off-bridge spectrum, which acts as a suppressor for the main frequency content of off-bridge vibrational sources, as shown in the following equation:

$$F[k] = \frac{1}{X_{\text{off}}[k]} \tag{1}$$

where X_{off} = discrete-time Fourier transform (DTFT) of the off-bridge acceleration signal; and F = filter shape. After applying this filter to the on-bridge data, the only source of vibration being absent from the off-bridge data, which is the bridge, is amplified through the following equation:

$$X^{\text{f}}_{\text{on}}[k] = F[k]X_{\text{on}}[k] = \frac{X_{\text{on}}[k]}{X_{\text{off}}[k]} \tag{2}$$

where X_{on} and X^{f}_{on} = unfiltered and filtered on-bridge spectra, respectively.

Among all factors presented in Fig. 1, the speed and road roughness are the most effective ones on the vehicle-recorded vibrations. Therefore, a database of off-bridge vibrations per different speed and roughness levels was proposed in a previous study (Shirzad-Ghaleroudkhani and Gül 2021). Thus, matching cases of on-bridge speed and roughness level are recalled from the off-bridge database in order to design the corresponding inverse filter. The flowchart of the inverse filtering method is presented in Fig. 2.

Mel-Frequency Cepstral Analysis

Cepstral analysis, as a powerful method in speech recognition and natural language processing, is defined as the inverse Fourier transform of the logarithm of the spectrum of a signal. MFC analysis is one of the methods employed in the cepstral environment. The outputs of Mel-frequency cepstral analysis are a series of features called Mel-frequency cepstral coefficients, herein MFCCs. These features represent important patterns in the frequency content of the signal, which is a major improvement compared to the peak detection. In addition, MFCCs are extracted with a higher focus on the lower frequencies, which are desired from the bridge health monitoring point of view.

The process of extracting MFCCs from acceleration signals in this study is presented in Fig. 3. After transforming the acceleration

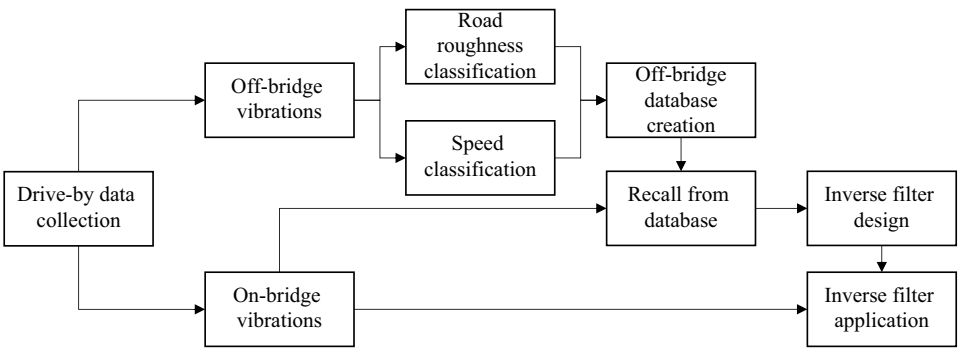


Fig. 2. Flowchart of inverse filtering method.

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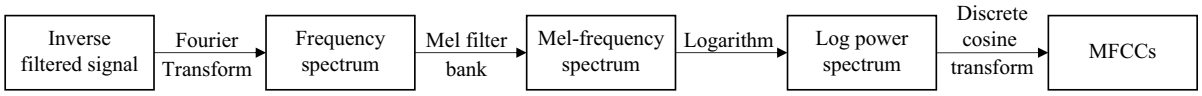


Fig. 3. The process of MFCC extraction.

signal to the frequency domain using the Fourier transform, the spectrum is transformed to a Mel-frequency environment. The relationship between the Mel-scale and Hertz-scale in speech recognition applications is designed based on the frequencies detectable by the human hearing system, which ranges between 2,000 and 5,000 Hz (Gelfand 1997). Since this frequency range is far from the major frequency content of bridge structures, an adapted version of the Mel-scale is proposed in the literature (Balsamo et al. 2014), as provided in the follow equations:

$$m = M(f) = 5\ln\left(1 + \frac{f}{5}\right)$$
$$f = M^{-1}(m) = 5(e^{m/5} - 1) \tag{3}$$

Eq. (3) shows relationships for converting the Hertz-scale, denoted by f, to the Mel-scale, denoted by m, and vice versa. The main purpose of this relationship is to relate evenly distributed Mel-frequencies with exponentially increasing intervals of Hertz-scale, leading to more focus on lower frequencies (Balsamo et al. 2014). The following equation locates those unevenly spaced points:

$$f_i = M^{-1}\left(M(f_1) + (i-1) \times \frac{M(f_n) - M(f_1)}{n-1}\right) \tag{4}$$

In Eq. (4), $f_i = i$th frequency selected unevenly from the Hertz-scale that corresponds to the ith evenly scaled Mel-scale frequency; and n = number of key frequencies selected from the frequency range of f_1–f_n. Note that the number of key frequencies will later contribute to the number of coefficients of MFCCs. This relationship is illustrated in Fig. 4 for 20 key frequencies in the range of 0–50 Hz. It can be seen that the frequency intervals are increasing exponentially in the Hertz-scale while being constant in the Mel-scale.

In the next step, Mel filter bank is built at the selected key frequencies through a series of triangular filter shapes (Rabiner and Schafer 2007) using the following equation:

$$H_i[k] = \begin{cases} 0 & k < f_{i-1} \\ \dfrac{2(k - f_{i-1})}{(f_{i+1} - f_{i-1})(f_i - f_{i-1})} & f_{i-1} \le k < f_i \\ \dfrac{2(f_{i+1} - k)}{(f_{i+1} - f_{i-1})(f_{i+1} - f_i)} & f_i \le k < f_{i+1} \\ 0 & k \ge f_{i+1} \end{cases}, \quad i = 2, 3, \ldots, n-1 \tag{5}$$

The function $H_i(k)$ in Eq. (5) represents the ith triangular filter bank at the corresponding key frequency. The filter bank plot corresponding to selected key frequencies in Fig. 4 is demonstrated in Fig. 5. It can be observed that the filter amplitude decreases with frequency, which is in accordance with prioritizing lower frequencies.

The filter bank is then applied to the frequency spectrum of the signal, calculating the total energy for each filter as follows:

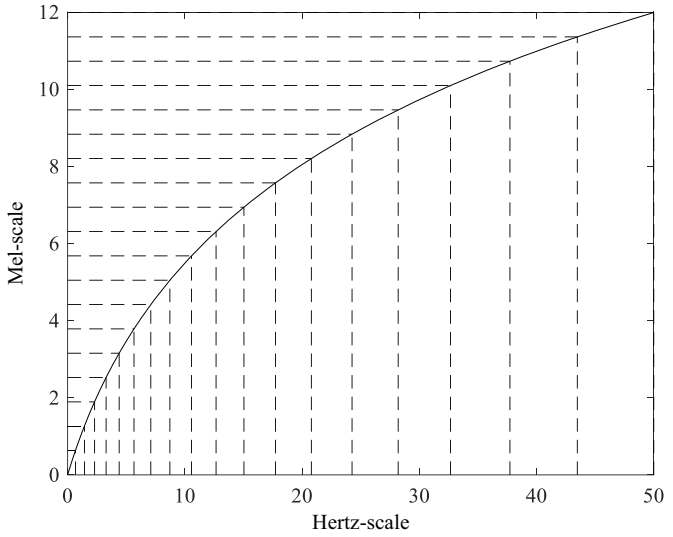


Fig. 4. Relationship between Hertz and Mel scale.

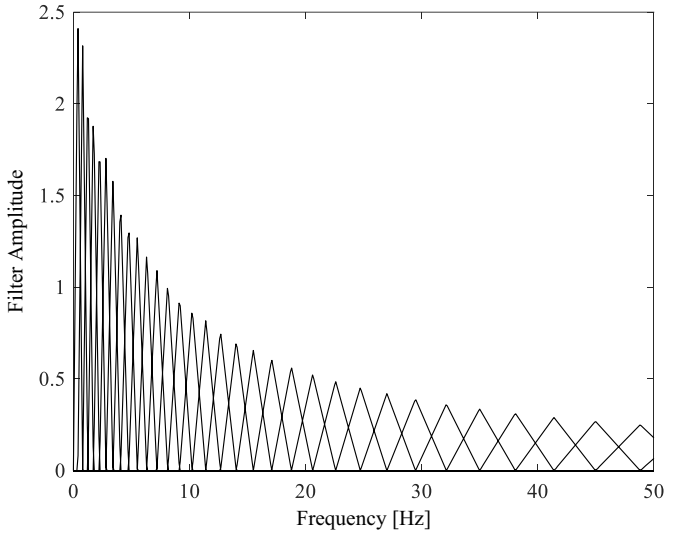


Fig. 5. Mel filter bank.

$$E[i] = \sum_k H_i[k] X_{\text{on}}^{\text{f}}[k], \quad i = 2, 3, \ldots, n-1 \tag{6}$$

where X_{on}^{f} = frequency spectrum of the on-bridge acceleration signal after inverse filtering according to the details provided in the previous section.

Following the process shown in Fig. 3, the next step is taking the logarithm of the total energy for each filter at each of the Mel-frequencies. Then, the discrete cosine transform of the logged energies will lead to MFCCs:

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$$C_j = \sum_{i=2}^{n-1} \log E[i] \cos\left[\frac{\pi}{\pi - 2}\left(i - \frac{3}{2}\right)j\right], \quad j = 1, 2, \ldots, n-2 \tag{7}$$

where $C_j = j$th coefficient of MFCCs. As mentioned before, the number of MFCCs depends on the number of filters, i.e., the number of key frequencies. The extracted MFCCs will then be used to create an abnormality index, which is explained in the following.

Abnormality Index

As mentioned before, MFCCs are expected to capture the pattern in a spectrum. Thus, major changes in the captured MFCCs could be a representation of a pattern change in the sources of vibrations. Without applying inverse filtering, it is expected that the captured pattern is mostly due to operational effects, i.e., vehicle and road parameters, rather than the bridge structures. However, after applying inverse filtering, it is expected that the presence of the bridge frequency content will be improved. Hence, the difference between the MFCCs of the two scenarios demonstrates a change in the frequency content of the main source, i.e., the bridge. Therefore, the Euclidean distance of the MFCCs is considered as follows:

$$\mathrm{DC_u} = \sqrt{\sum_{j=n_1}^{n_2} (C_{j,u} - C_{j,b})^2} \tag{8}$$

In Eq. (8), $\mathrm{DC_u}$ = "Difference of Coefficients" for the unknown case; $C_{j,\mathrm{b}} = j$th coefficient of the baseline case; and $C_{j,\mathrm{u}} = j$th coefficient of the unknown case. Practically, only some of them are selected for further analysis, which are shown by the range of n_1–n_2 in Eq. (8).

Ideally, if the inverse filtering was capable of removing 100% of the operational effects from the drive-by spectrum, the DC would be zero considering the healthy bridge as the unknown case. However, due to errors in the estimation of the surface roughness and the effect of continuously varying speed of the vehicle, the elimination of operational effects is not completely perfect and the DC will not get theoretically zero. In order to account for the remaining effects of the operational factor, the distribution of the DCs calculated from the passage of multiple vehicles over the bridge is considered. In other words, the change in the distribution of DCs of vehicles passing over a bridge would be an index for an abnormality in the bridge structure. A sample of the distribution of DC parameters for a real-life case study is presented in Fig. 6.

Quantification of the distribution comparison is achieved using the Z test through the following formula:

$$\mathrm{Z} = \frac{|\mu_2 - \mu_1|}{\sqrt{\sigma_2^2 + \sigma_1^2}} \tag{9}$$

where μ_1 and μ_2 = means; and σ_1 and σ_2 = standard deviations of the fitted normal distributions of DC for two different scenarios. In fact, the Z test will show how the distribution of DCs has changed between two specific points in time and hence will be denoted as the abnormality index in this study. The application of the Z test as a damage indicator in SHM through comparing distributions of extracted features has been previously studied (Moustafa et al. 2010; Ye et al. 2014; Zhang et al. 2022) in the literature. In order to assess the capability of the proposed method for detecting changes in bridge state, first, a series of laboratory and real-life experiments are conducted with the focus on the elimination of operational effects. Later, the performance of the method for damage detection is investigated.

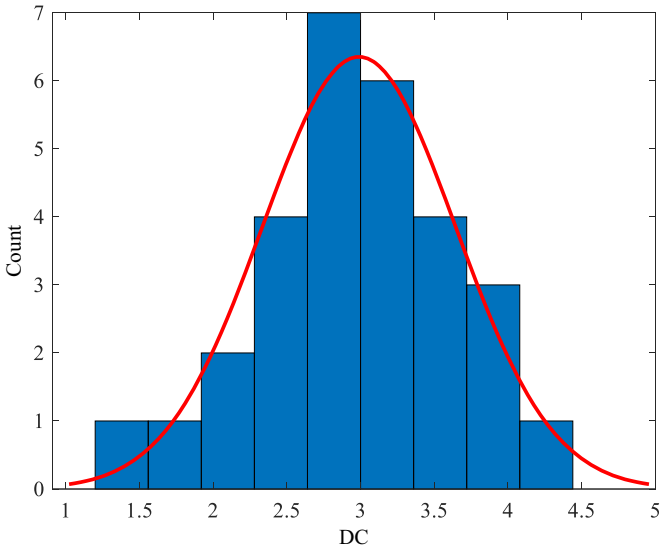


Fig. 6. A sample of the distribution of DC parameters for a real-life case study (on the High-Level Bridge discussed in later sections).

Application of the Proposed Methodology for Eliminating Operational Effects

This section focuses on the application of the proposed methodology in eliminating operational effects from indirectly recorded vibrations. As mentioned, vehicle features, including speed, suspension system, and engine vibrations, as well as road roughness significantly dominate the vibrations recorded on the passing vehicle. The performance of the proposed approach in eliminating such effects is assessed in the following sections using first laboratory experiments and then real-life experiments.

Laboratory Experiments

Experimental Setup

In order to investigate the capability of the methodology in suppressing operational effects, controlled laboratory experiments are

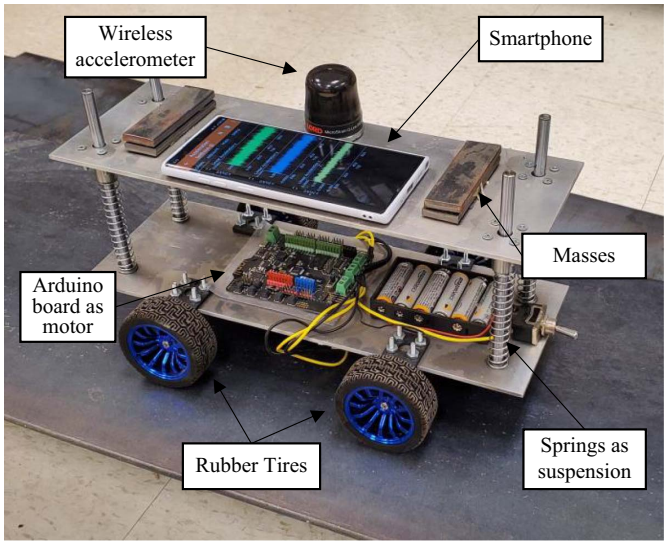


Fig. 7. The custom–designed and –built robot car used in the experiment.

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Fig. 8. Bridge model employed in the experimental study.

Table 1. Features of different elements of the experimental setup

Element	Parameter	Value
Robot car	Plate dimensions (mm)	$350 \times 125 \times 3.1$
	Total weight (kg)	2.3
	Top plate weight (kg)	1.3
	Spring coefficient (N/m)	425, 615, 726
	Speed (m/s)	0.2, 0.3, 0.4
Bridge	Steel grade	W44
	Modulus of elasticity (GPa)	200
	Yield strength (MPa)	250
	Deck dimensions (mm)	$2{,}000 \times 330 \times 6.35$
	Deck weight (kg)	30

conducted. The setup includes a custom-designed and -built robot car capable of maintaining different suspension levels and speed values in order to represent various vehicles passing over a bridge, as shown in Fig. 7. Additionally, a steel plate resting on a pin-roller support system is used as the bridge model, presented in Fig. 8. Note that the off-bridge simulation required for the design of the inverse filter was conducted using the robot car moving on the same steel plate resting on the floor, as the effect of the road roughness change was not the focus of this controlled experiment. Furthermore, the features of the robot car and the bridge are presented in Table 1. As seen, three different suspension levels and three different speed values yield nine unique combinations of vehicle features, and the results are presented in the following section.

Results

In this section, the abnormality index explained in the Methodology Section is calculated for four trial series of robot car travels over the bridge considering a unique baseline case. Each trial consists of three travels for nine unique combinations of car features, resulting in 108 robot car travels in total. Note that at this point of study, the bridge is undamaged for the unknown trials as the focus is on assessing the elimination of operational effects only. In other words, the baseline and unknown cases for comparing MFCCs and calculating the abnormality index are both the undamaged bridge.

Fig. 9 illustrates the fitted normal distributions for DC parameters of two separate trials. Fig. 9(a) shows the distributions without filtering, while Fig. 9(b) shows the results with inverse filtering. As seen, there is a slight difference in the distribution of DC parameters while the filter has not been applied, which is minimized after applying inverse filtering. In addition, the distributions have been narrower in the filtered versions as the presence of operational effects has been reduced. Note that the experiment has been conducted in controlled ideal conditions, such as similar surface roughness levels, and these differences are expected to be more significant under real-life conditions, as discussed in the next part of this section.

In order to quantify the distribution comparison, the abnormality index calculations are presented in Table 2 for all six possible scenarios out of four trials. In fact, the index for each scenario is calculated by performing the Z test between the distributions of two different trials for the healthy bridge. As seen, the inverse filter significantly reduces the abnormality index, 89% on average, which is expected due to the elimination of operational effects, including the suspension and speed effects. It is shown in the Damage Detection Section that the proposed inverse filtering-based methodology and abnormality index are very effective for detecting bridge damage using indirect monitoring under operational effects.

Real-Life Experiments

Experimental Setup

In this section, the performance of the proposed method in eliminating operational effects is investigated in real-life conditions.

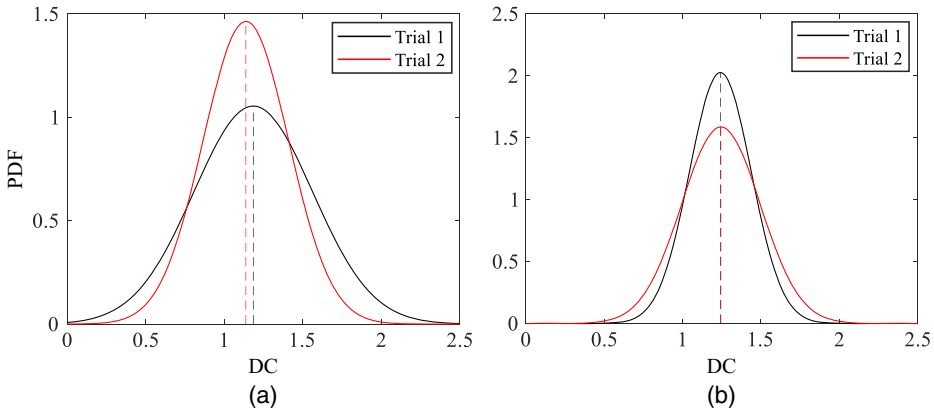


Fig. 9. Fitted normal distribution of DC parameter for two cases (a) without filtering; and (b) with inverse filtering.

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Five different bridges in the city of Edmonton, Alberta, Canada, are studied as detailed in Table 3. All bridges span over North Saskatchewan River and their locations are shown in Fig. 10. In order to account for the effects of vehicle and device types, four different smartphones mounted on four different vehicles were employed in this study, which details are presented in Table 4. The pictures of two cars as well as a sample of the placement of the smartphone are shown in Fig. 11. As seen, the vehicles are a combination of Sedan and SUV types, and smartphones are from different models, with accelerometers of different maximum sampling frequency of 200–500 Hz.

Table 2. Abnormality index for trial cases in the healthy bridge condition

Scenario	Unfiltered	Inverse filtered
1	0.11	0.038
2	0.26	0.023
3	0.09	0.008
4	0.13	0.020
5	0.37	0.037
6	0.48	0.037
Average	0.24	0.027

Table 3. Details of the bridges studied in the real-life experiment

Bridge name	Length (m)	System	Year built	Last rehabilitation
Groat Bridge	315	Steel girder	1955	2020
High-Level Bridge	777	Steel truss	1913	1995
Walterdale Bridge	214	Steel through-arch	2017	N/A
Low-Level Bridge	213	Steel truss	1900	2006
Dawson Bridge	236	Steel truss	1912	2010

Results

In the experiments, all the vehicles traveled each bridge twice. In comparison to the laboratory experiments where more data were available, in real-life experiments, due to the limited amount of data, the considered baseline case is the direct bridge measurements. In fact, fixed accelerometers were mounted on each bridge and their acceleration data were used for the baseline case. Later, with the collection of drive-by data from vehicles, the abnormality index is calculated compared to the baseline data. This way, it would be possible to concentrate on the elimination of operational factors, as the baseline case is direct bridge vibrations and contains no operational effects.

Table 4. Vehicle and smartphone devices employed in the real-life experiment

Vehicle model	Vehicle type	Smartphone model	Maximum sampling frequency (Hz)
Honda Civic (1)	Sedan	Samsung Galaxy Note 10+	400
Honda Civic (2)	Sedan	BlackBerry Priv	200
Honda Pilot	SUV	Huawei Mate 20 Pro	500
Toyota Rav4	SUV	Samsung Galaxy S10	415

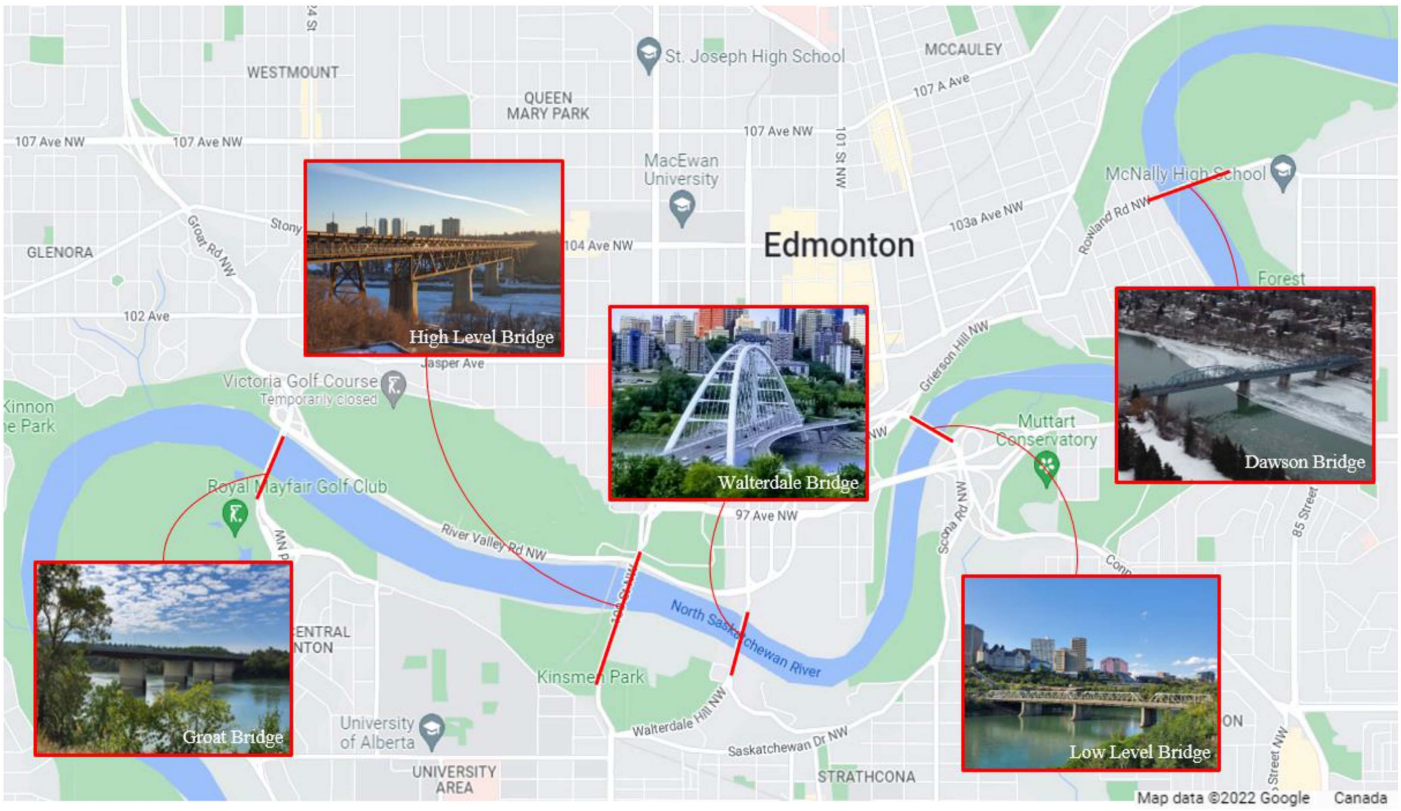


Fig. 10. Locations and images of the bridges studied in the real-life experiment. (Images by Mustafa Gül.) (Map data © 2022 Google.) [Reprinted from Elsevier Books, Nima Shirzad-Ghaleroudkhani, Qipei Mei, Mustafa Gül, "The Rise of Smart Cities," pp. 609–624 © 2022, with permission from Elsevier.]"

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Fig. 11. Pictures of two of the vehicles as well as smartphone placement inside a vehicle. (Images by Mustafa Gül.)

Each case of this study includes the data extracted from one sedan and one SUV vehicle passing over each bridge. The distribution of DC parameters for each case is shown in Fig. 12. The figure consists of five rows, presenting results for each bridge, and two columns. In column (a), the boxplot presentation of the distributions is provided, which is the five-number summary of each case, i.e., the minimum, first quartile, median, third quartile, and maximum. Each plot in column (a) shows four distributions, two data cases denoted by C1 and C2, and unfiltered versus inverse-filtered results denoted by (U) and (F), respectively. Column (b) presents the fitted normal distribution for DC parameters for each case, where black curves represent distributions using unfiltered data and red curves represent those of inverse filtered. As illustrated in Fig. 12, without applying the inverse filtering method, for almost all bridges studied here, the change in the distribution of DC parameters is significant. Since the MFCCs used in the calculation of DCs are the representers of the patterns in the spectrum of a signal, these substantial changes in the frequency content could result in a false abnormality detection of the bridge, which is in fact due to the effect of the operational factors. However, since the bridges were not damaged during this study, it is expected that a robust method of filtering out operational effects produces similar distributions, which is seen in Fig. 12. Moreover, inverse filtering notably reduces the DC values in comparison to the unfiltered cases, further corroborating the fact that undesired operational effects are less significant in the inverse-filtered data. It is noteworthy that the level of the bridge excitation and the amount of vibrational data availability, which depend on the bridge length and structure type, may affect the presence of bridge vibrations in the mixed accelerations recorded on the vehicle. For instance, High-Level Bridge in this study has both the highest length and the most excitation level among all studied bridges. Therefore, as expected, the presence of the bridge vibrations among unfiltered data is significantly higher than in other cases, which is resonated in Fig. 12. As seen, the shift in the DC values is much less compared to other bridges, even in the unfiltered case.

In order to quantify the results, the proposed abnormality index is calculated for this study and presented in Table 5. As expected, the abnormality index is significantly reduced using the inverse filtering method. In addition, the order of abnormality index using inverse filtering in a real-life study is comparable to the laboratory experiment, i.e., order of magnitude of 10^{-2}, which shows the successful performance of the method in eliminating operational effects even in real-life conditions with challenging factors compared to a controlled laboratory environment. Using the results provided in this section, it can be concluded that without applying the inverse filtering method, the MFC analysis is not capable of capturing patterns related to bridge dynamics in the mixed drive-by vibrations, as the bridge features are barely detectable in the significant presence of vehicle and road roughness effects under real-life conditions.

Damage Detection

This section covers the application of the proposed methodology for bridge damage detection. Since there was no real-life data on a damaged bridge available, the damage detection study was only investigated through a laboratory experiment. The experimental setup is similar to the previous section. However, in addition to the healthy bridge case, four damaged cases are employed. The damage in the bridge is modeled using the reduction in the cross-sectional area of the bridge deck at different locations. The details and pictures of all damage cases are presented in Table 6 and Fig. 13, respectively. The damage cases are designed so that both the effect of the damage severity and location are considered. Note that the steel plates used for all damage cases are similar in material and dimensions, except for the cross-sectional area reduction.

The distribution of DC parameters for damage cases at the center and quarter of the bridge length are presented in Figs. 14 and 15, respectively. The rows in those figures represent results

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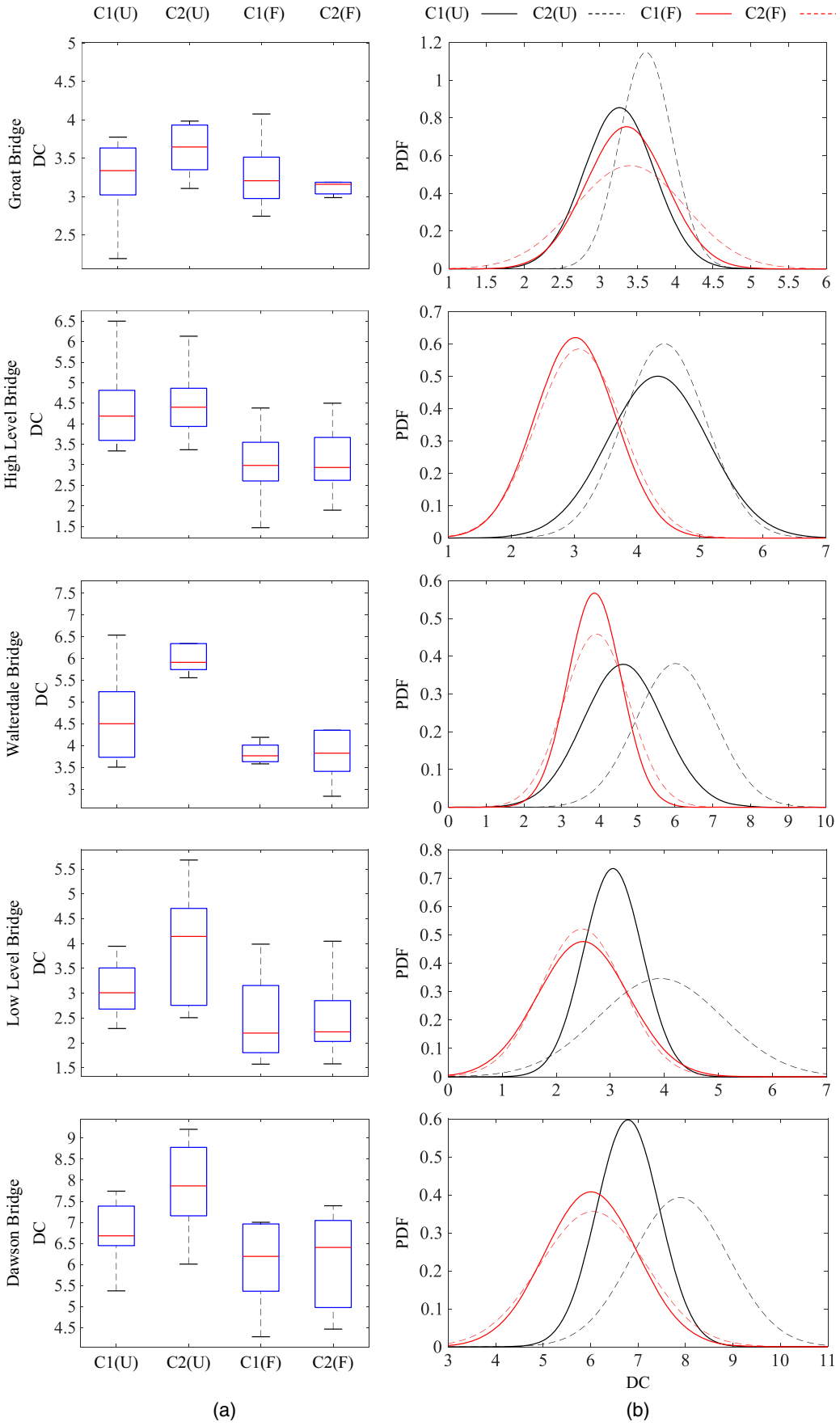


Fig. 12. Distribution of DC parameters for each case (C1, C2) without filtering (U) and using inverse filtering method (F) plotted using (a) boxplots; and (b) fitted normal distributions.

Table 5. Abnormality index for real-life study

Bridge	Unfiltered	Inverse filtered
Groat Road	0.606	0.056
High Level	0.099	0.061
Walterdale	0.936	0.043
MacDonald	0.688	0.015
Low Level	0.905	0.027

Table 6. Details of the bridge cases in the laboratory experiment for damage detection analyses

Damage code	Damage location	Cross-sectional area reduction
B	No damage (Baseline)	N/A
C1	Center	15%
C2	Center	30%
Q1	Quarter	15%
Q2	Quarter	30%

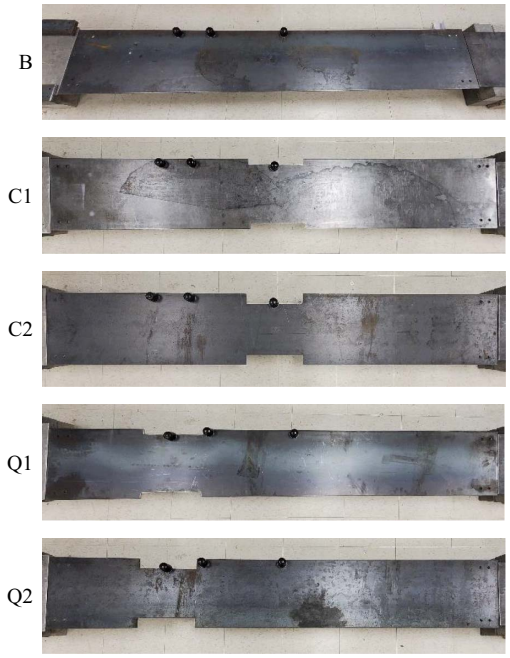


Fig. 13. Images of damage cases employed in the laboratory experiment.

using unfiltered versus inverse-filtered signals, while the columns, similar to Fig. 12, illustrate the boxplot and fitted normal distributions of the DC parameter. Note that each figure contains one distribution for the baseline case, i.e., healthy bridge, which will be used as a benchmark for damage detection. As seen in the first rows of Figs. 14 and 15, it is barely possible to detect a shift in the distribution of the DC parameter, representing a change in the pattern of the frequency spectrum. It is noteworthy that such small shifts can occur in the healthy conditions of the bridge according to the results seen in the previous section. However, the second rows of Figs. 14 and 15 demonstrate a considerable shift between damage cases, which increases with the severity of the damage, substantiating the effect of inverse filtering for improving damage detection accuracy.

Quantification of the damage detection performance is conducted using the proposed abnormality index, which results are presented in Table 7. The abnormality indices are calculated by comparing the distribution of the DC parameters in the healthy baseline case with each damage case. Although the indices increase with damage severity in the unfiltered analysis, comparing the index values with those of the healthy cases in Table 2 shows that detecting damage on the bridge is almost impossible. In fact, the abnormality index shows similar levels in healthy and damaged cases without an appropriate filtering method, which supports the notion that the drive-by vibrations are dominated by operational factors. On the other hand, inverse-filtered indices are significantly larger in the damaged cases compared to the healthy cases in Table 2. Therefore, not only the inverse filtering decrease abnormality indices for healthy cases, but it also increases those of the damaged cases, yielding far more accurate damage detection. Although the damage levels considered in this experiment are substantial and do not mimic damage levels in the in-service bridge, it was shown that operational effects in indirect bridge monitoring are significant enough to overshadow such large damage levels, and an appropriate filtering method would be necessary to detect them.

It is worth noting that since the proposed damage detection method relies on the recognition of changes in the spectral pattern of vibrations, any structural damage that affects the spectral pattern should in theory be detectable with the proposed method. Thus, other damage types that may occur beyond the bridge deck, e.g., on the substructure or other components of the superstructure, are also deemed to be identifiable. It is expected that any changes in major components of the complex system of the bridge structure will affect the spectral contents of its vibrations. Consequently, major damages that occur on the substructure, such as pillars or even column foundations, may act as boundary condition changes and thus affect the vibrational components recorded on the deck. Such changes could then be extracted by detecting pattern changes in the spectrum using the proposed method.

Conclusion

This paper proposes a bridge monitoring method considering the operational effect using drive-by vibrations recorded on passengers' smartphones. Since drive-by data are significantly dominated by operational factors, including engine vibrations, suspension system, road roughness, a recently developed inverse filtering method is employed to suppress such factors and magnify bridge dynamic features. In addition, a MFC analysis is employed to capture patterns in the frequency spectrum instead of looking for peaks. An abnormality index is proposed, representing a change in the patterns inside the frequency spectrum, that would be interpreted as a change in the health state of the bridge structure.

The performance of the proposed method was first assessed through controlled laboratory experiments, where a robot car modeled the vehicle and a steel plate resting on a pin-roller boundary condition represented the bridge. Different combinations of suspension stiffness and speed values were considered to account for a variety of vehicles passing over the bridge using the robot car. It was shown that the frequency content of the spectrum would be overshadowed by operational factors without applying inverse filtering. Similar conclusions were made in a real-life study conducted on five major bridges in the city of Edmonton. Four different smartphones mounted on four different vehicles were employed to mimic different device-vehicle combinations in a crowdsensing framework. The main deduction was the importance of applying

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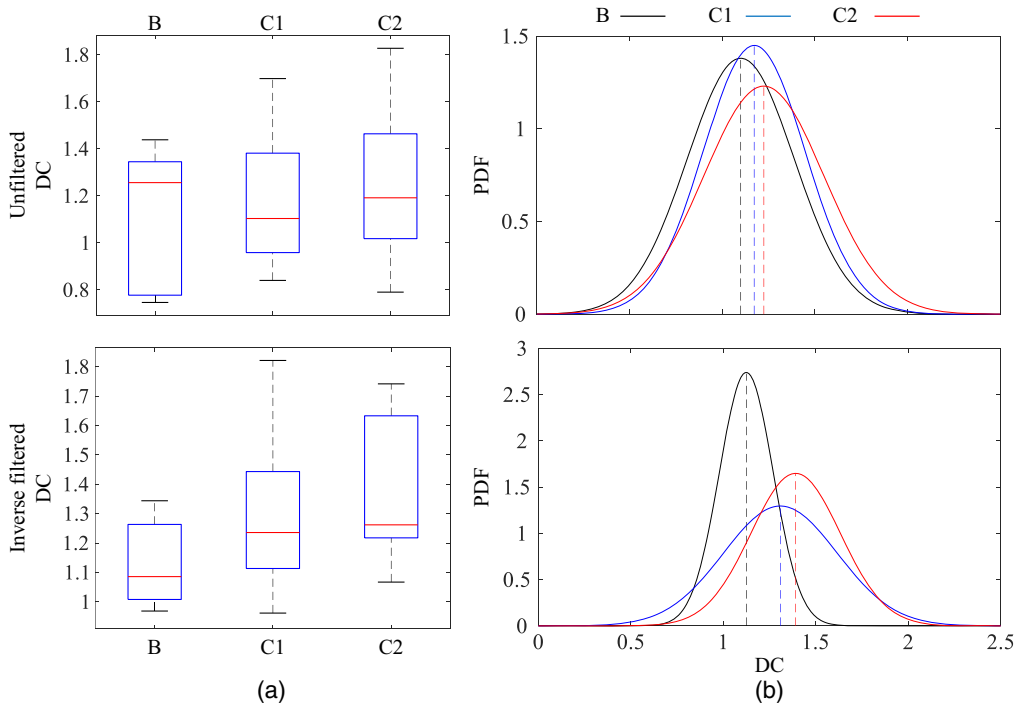


Fig. 14. Distribution of DC parameters for the baseline and damage cases at the center (B, C1, and C2) plotted using (a) boxplots; and (b) fitted normal distributions.

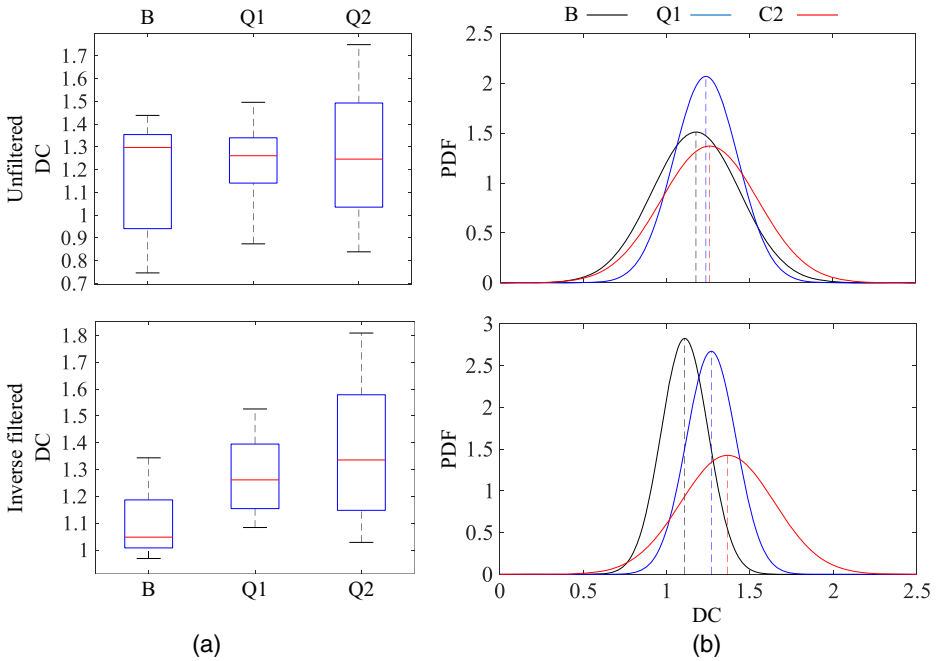


Fig. 15. Distribution of DC parameters for the baseline and damage cases at the quarter (B, Q1, and Q2) plotted using (a) boxplots; and (b) fitted normal distributions.

Table 7. Abnormality index for the damage detection study

Damage case	Unfiltered	Inverse filtered
C1	0.18	0.54
C2	0.29	0.94
Q1	0.18	0.78
Q2	0.21	0.82

inverse filtering for detecting patterns related to bridge vibrations in a spectrum.

The laboratory experiments were then extended to investigate the damage detection capability of the proposed method. Four damage cases were considered through two levels of cross-sectional area reduction at the center and quarter length of the bridge deck. The proposed abnormality index was capable of detecting the

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damage in all cases and was sensitive to the severity of the damage. Comparison of the abnormality index between healthy and damaged cases corroborated the fact that without an appropriate filtering method, the index results in similar levels for both healthy and damaged conditions, substantiating the dominancy of operational effects on the drive-by vibrations.

Considering the significant performance of the proposed method in eliminating operational effects and detecting damage on the bridge, it is crucial to indicate that other factors need to be implemented in future studies. The underlying assumption in this study was the stationary position of the smartphone inside the vehicle. Hence, potential passenger's interactions and motions while using smartphones are other operational factors that need to be addressed in future studies. In addition, environmental factors, including temperature, humidity, and wind, are among the major features challenging bridge monitoring in real-life conditions, including drive-by methods. Such factors need to be considered in the future. The method can then be implemented in a crowdsensing framework, analyzing the vibrations recorded on passengers' smartphones to detect the state of a population of bridges in large geographic areas.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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References

Alexandre, P., and P. Lockwood. 1993. "Root cepstral analysis: A unified view. Application to speech processing in car noise environments." *Speech Commun.* 12 (3): 277–288. https://doi.org/10.1016/0167-6393(93)90099-7.

Andrade, C., C. Alonso, and J. Sarŕa. 2002. "Corrosion rate evolution in concrete structures exposed to the atmosphere." *Cem. Concr. Compos.* 24 (1): 55–64. https://doi.org/10.1016/S0958-9465(01)00026-9.

Arora, V., S. P. Singh, and T. K. Kundra. 2009. "Damped model updating using complex updating parameters." *J. Sound Vib.* 320 (1–2): 438–451. https://doi.org/10.1016/j.jsv.2008.08.014.

ASCE. 2021. *Report card for America's infrastructure*. Reston, VA: ASCE.

Balsamo, L., R. Betti, and H. Beigi. 2014. "A structural health monitoring strategy using cepstral features." *J. Sound Vib.* 333 (19): 4526–4542. https://doi.org/10.1016/j.jsv.2014.04.062.

Childers, D. G., D. P. Skinner, and R. C. Kemerait. 1977. "The cepstrum: A guide to processing." *Proc. IEEE* 65 (10): 1428–1443. https://doi.org/10.1109/PROC.1977.10747.

CIRC (Canada Infrastructure Report Card). 2019. *Canada infrastructure report card 2019*. Toronto: Canadian Consulting Engineer.

Elhattab, A. 2017. "Drive-by bridge damage monitoring: Concise review." *Civ. Eng. Res. J.* 1 (1): 25–30. https://doi.org/10.19080/CERJ.2017.01.555555.

Furui, S. 1981. "Cepstral analysis technique for automatic speaker verification." *IEEE Trans. Acoust. Speech Signal Process.* 29 (2): 254–272. https://doi.org/10.1109/TASSP.1981.1163530.

Gelfand, S. A. 1997. *Essentials of audiology*. New York: Thieme.

Guettala, A., and A. Abibsi. 2006. "Corrosion degradation and repair of a concrete bridge." *Mater. Struct.* 39 (4): 471–478. https://doi.org/10.1007/s11527-005-9046-z.

Hsieh, K. H., M. W. Halling, and P. J. Barr. 2006. "Overview of vibrational structural health monitoring with representative case studies." *J. Bridge Eng.* 11 (6): 707–715. https://doi.org/10.1061/(ASCE)1084-0702(2006)11:6(707).

Huth, O., G. Feltrin, J. Maeck, N. Kilic, and M. Motavalli. 2005. "Damage identification using modal data: Experiences on a prestressed concrete bridge." *J. Struct. Eng.* 131 (12): 1898–1910. https://doi.org/10.1061/(ASCE)0733-9445(2005)131:12(1898).

Keenahan, J., E. J. OBrien, P. J. McGetrick, and A. Gonzalez. 2013. "The use of a dynamic truck–trailer drive-by system to monitor bridge damping." *Struct. Health Monit.* 13 (2): 143–157. https://doi.org/10.1177/1475921713513974.

Khorram, A., F. Bakhtiari-Nejad, and M. Rezaeian. 2012. "Comparison studies between two wavelet based crack detection methods of a beam subjected to a moving load." *Int. J. Eng. Sci.* 51 (Apr): 204–215. https://doi.org/10.1016/j.ijengsci.2011.10.001.

Lederman, G., Z. Wang, J. Bielak, H. Noh, J. H. Garrett, S. Chen, J. Kovačević, F. Cerda, and P. Rizzo. 2014. "Damage quantification and localization algorithms for indirect SHM of bridges." In *Proc., 7th Int. Conf. of Bridge Maintenance, Safety and Management, IABMAS 2014*, 640–647. Abingdon, UK: Routledge.

Li, W. M., Z. H. Jiang, T. L. Wang, and H. P. Zhu. 2014. "Optimization method based on generalized pattern search algorithm to identify bridge parameters indirectly by a passing vehicle." *J. Sound Vib.* 333 (2): 364–380. https://doi.org/10.1016/j.jsv.2013.08.021.

Lin, C. W., and Y. B. Yang. 2005. "Use of a passing vehicle to scan the fundamental bridge frequencies: An experimental verification." *Eng. Struct.* 27 (13): 1865–1878. https://doi.org/10.1016/j.engstruct.2005.06.016.

Liu, J., M. Bergés, J. Bielak, J. H. Garrett, J. Kovačević, and H. Y. Noh. 2019. "A damage localization and quantification algorithm for indirect structural health monitoring of bridges using multi-task learning." *AIP Conf. Proc.* 2102 (1): 090003. https://doi.org/10.1063/1.5099821.

Liu, T., and R. W. Weyers. 1998. "Modeling the dynamic corrosion process in chloride contaminated concrete structures." *Cem. Concr. Res.* 28 (3): 365–379. https://doi.org/10.1016/S0008-8846(98)00259-2.

Malekjafarian, A., P. J. McGetrick, and E. J. Obrien. 2015. "A review of indirect bridge monitoring using passing vehicles." *Shock Vib.* 2015 (Aug): 1. https://doi.org/10.1155/2015/286139.

Malekjafarian, A., and E. J. O'Brien. 2014a. "Application of output-only modal method in monitoring of bridges using an instrumented vehicle." In *Civil engineering research in Ireland*. Dublin, Ireland: Civil Engineering Research Association of Ireland.

Malekjafarian, A., and E. J. OBrien. 2014b. "Identification of bridge mode shapes using short time frequency domain decomposition of the responses measured in a passing vehicle." *Eng. Struct.* 81 (May): 386–397. https://doi.org/10.1016/j.engstruct.2014.10.007.

Matarazzo, T. J., P. Santi, S. N. Pakzad, K. Carter, C. Ratti, B. Moaveni, C. Osgood, and N. Jacob. 2018. "Crowdsensing framework for monitoring bridge vibrations using moving smartphones." *Proc. IEEE* 106 (4): 577–593. https://doi.org/10.1109/JPROC.2018.2808759.

McGetrick, P. J., and C. W. Kim. 2013. "A parametric study of a drive by bridge inspection system based on the Morlet wavelet." *Key Eng. Mater.* 569–570 (Aug): 262–269. https://doi.org/10.4028/www.scientific.net/KEM.569-570.262.

McGetrick, P. J., and C. W. Kim. 2014. "An indirect bridge inspection method incorporating a wavelet-based damage indicator and pattern recognition." In *Proc., Int. Conf. on Structural Dynamic, EURODYN*, 2605–2612. Porto, Portugal: European Association for Structural Dynamics.

Mei, Q., and M. Gül. 2019. "A crowdsourcing-based methodology using smartphones for bridge health monitoring." *Struct. Health Monit.* 18 (5–6): 1602–1619. https://doi.org/10.1177/1475921718815457.

Mei, Q., M. Gül, and M. Boay. 2019. "Indirect health monitoring of bridges using Mel-frequency cepstral coefficients and principal component analysis." *Mech. Syst. Signal Process.* 119 (Sep): 523–546. https://doi.org/10.1016/j.ymssp.2018.10.006.

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Mei, Q., M. Gül, and N. Shirzad-Ghaleroudkhani. 2020. "Towards smart cities: Crowdsensing-based monitoring of transportation infrastructure using in-traffic vehicles." *J. Civ. Struct. Health Monit.* 10 (4): 653–665. https://doi.org/10.1007/s13349-020-00411-6.

Mei, Q., N. Shirzad-Ghaleroudkhani, M. Gül, S. F. Ghahari, and E. Taciroglu. 2021. "Bridge mode shape identification using moving vehicles at traffic speeds through non-parametric sparse matrix completion." *Struct. Control Health Monit.* 28 (7): e2747. https://doi.org/10.1002/stc.2747.

Moustafa, A., S. Mahadevan, M. Daigle, and G. Biswas. 2010. "Structural and sensor damage identification using the bond graph approach." *Struct. Control Health Monit.* 17 (2): 178–197. https://doi.org/10.1002/stc.285.

Nazarian, E., F. Ansari, M. Asce, X. Zhang, and T. Taylor. 2016. "Detection of tension loss in cables of cable-stayed bridges by distributed monitoring of bridge deck strains." *J. Struct. Eng.* 142 (6): 04016018. https://doi.org/10.1061/(ASCE)ST.1943-541X.0001463.

OBrien, E. J., P. J. Mcgetrick, and A. González. 2014. "A drive-by inspection system via vehicle moving force identification." *Smart Struct. Syst.* 13 (5): 821–848. https://doi.org/10.12989/sss.2014.13.5.821.

On, C. K., P. M. Pandiyan, S. Yaacob, and A. Saudi. 2006. "Mel-frequency cepstral coefficient analysis in speech recognition." In *Proc., Int. Conf. on Computing and Informatics*, 1–5. New York: IEEE. https://doi.org/10.1109/ICOCI.2006.5276486.

Oshima, Y., K. Yamamoto, and K. Sugiura. 2014. "Damage assessment of a bridge based on mode shapes estimated by responses of passing vehicles." *Smart Struct. Syst.* 13 (5): 731–753. https://doi.org/10.12989/sss.2014.13.5.731.

Pandey, A. K., M. Biswas, and M. M. Samman. 1991. "Damage detection from changes in curvature mode shapes." *J. Sound Vib.* 145 (2): 321–332. https://doi.org/10.1016/0022-460X(91)90595-B.

Rabiner, L. R., and R. W. Schafer. 2007. "Introduction to digital speech processing." *Found. Trends Signal Process.* 1 (1–2): 1–194. https://doi.org/10.1561/2000000001.

Ramadan, O. E., and V. P. Sisiopiku. 2016. "Evaluation of merge control strategies at interstate work zones under peak and off-peak traffic conditions." *J. Transp. Technol.* 06 (03): 118–130. https://doi.org/10.4236/jtts.2016.63011.

Santini Bell, E., A. M. Asce, P. J. Lefebvre, M. Sanayei, M. Asce, B. Brenner, F. Asce, J. D. Sipple, and J. Peddle. 2013. "Objective load rating of a steel-girder bridge using structural modeling and health monitoring." *J. Struct. Eng.* 139 (10): 1771–1779. https://doi.org/10.1061/(ASCE)ST.1943-541X.0000599.

Shirzad-Ghaleroudkhani, N., and M. Gül. 2020. "Inverse filtering for frequency identification of bridges using smartphones in passing vehicles: Fundamental developments and laboratory verifications." *Sensors* 20 (4): 1190. https://doi.org/10.3390/s20041190.

Shirzad-Ghaleroudkhani, N., and M. Gül. 2021. "An enhanced inverse filtering methodology for drive-by frequency identification of bridges using smartphones in real-life conditions." *Smart Cities* 4 (2): 499–513. https://doi.org/10.3390/smartcities4020026.

Shirzad-Ghaleroudkhani, N., Q. Mei, and M. Gül. 2020. "Frequency identification of bridges using smartphones on vehicles with variable features." *J. Bridge Eng.* 25 (7): 04020041. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001565.

Shirzad-Ghaleroudkhani, N., Q. Mei, and M. Gül. 2022. "A crowdsensing-based platform for transportation infrastructure monitoring and management in smart cities." In *The rise of smart cities*, 609–624. Amsterdam, Netherlands: Elsevier.

Sitton, J. D., D. Rajan, and B. A. Story. 2020a. "Bridge frequency estimation strategies using smartphones." *J. Civ. Struct. Health Monit.* 10 (3): 513–526. https://doi.org/10.1007/s13349-020-00399-z.

Sitton, J. D., Y. Zeinali, D. Rajan, and B. A. Story. 2020b. "Frequency estimation on two-span continuous bridges using dynamic responses of passing vehicles." *J. Eng. Mech.* 146 (1): 04019115. https://doi.org/10.1061/(ASCE)EM.1943-7889.0001698.

Tokuda, K., T. Kobayashi, and S. Imai. 1995. "Adaptive cepstral analysis of speech." *IEEE Trans. Speech Audio Process.* 3 (6): 481–489. https://doi.org/10.1109/89.482216.

Vergin, R., D. O'Shaughnessy, and A. Farhat. 1999. "Generalized Mel frequency cepstral coefficients for large-vocabulary speaker-independent continuous-speech recognition." *IEEE Trans. Speech Audio Process.* 7 (5): 525–532. https://doi.org/10.1109/89.784104.

Wenzel, H. 2009. *Health monitoring of bridges*. New York: Wiley.

Xie, H., Y. Wang, and M. Liu. 2011. "Influences of climate change on reliability of bridge superstructure." *Appl. Mech. Mater.* 94–96 (Feb): 1549–1555. https://doi.org/10.4028/www.scientific.net/AMM.94-96.1549.

Yang, Y. B., and K. C. Chang. 2009. "Extraction of bridge frequencies from the dynamic response of a passing vehicle enhanced by the EMD technique." *J. Sound Vib.* 322 (4–5): 718–739. https://doi.org/10.1016/j.jsv.2008.11.028.

Yang, Y. B., Y. C. Li, and K. C. Chang. 2014. "Constructing the mode shapes of a bridge from a passing vehicle: A theoretical study." *Smart Struct. Syst.* 13 (5): 797–819. https://doi.org/10.12989/sss.2014.13.5.797.

Yang, Y. B., C. W. Lin, and J. D. Yau. 2004. "Extracting bridge frequencies from the dynamic response of a passing vehicle." *J. Sound Vib.* 272 (3–5): 471–493. https://doi.org/10.1016/S0022-460X(03)00378-X.

Yang, Y. B., and J. P. Yang. 2018. "State-of-the-art review on modal identification and damage detection of bridges by moving test vehicles." *Int. J. Struct. Stab. Dyn.* 18 (2): 1850025. https://doi.org/10.1142/S0219455418500256.

Ye, X., Q. Qin, M. Liu, J. Wang, and J. Wang. 2014. "Building damage detection from post-quake remote sensing image based on fuzzy reasoning." In *Proc., Int. Geoscience and Remote Sensing Symp.*, 529–532. New York: IEEE.

Yin, S. H., and C. Y. Tang. 2011. "Identifying cable tension loss and deck damage in a cable-stayed bridge using a moving vehicle." *J. Vib. Acoust.* 133 (2): 021007. https://doi.org/10.1115/1.4002128.

Zhang, G., R. S. Harichandran, and P. Ramuhalli. 2011. "Application of noise cancelling and damage detection algorithms in NDE of concrete bridge decks using impact signals." *J. Nondestr. Eval.* 30 (4): 259–272. https://doi.org/10.1007/s10921-011-0114-8.

Zhang, X., Y. Ding, H. Zhao, and L. Yi. 2022. "Long-term bridge performance assessment using clustering and Bayesian linear regression for vehicle load and strain mapping model." *Struct. Control Health Monit.* 29 (12): e3118. https://doi.org/10.1002/stc.3118.

Zhang, Y., S. T. Lie, and Z. Xiang. 2013. "Damage detection method based on operating deflection shape curvature extracted from dynamic response of a passing vehicle." *Mech. Syst. Signal Process.* 35 (1–2): 238–254. https://doi.org/10.1016/j.ymssp.2012.10.002.

Zhang, Y., L. Wang, and Z. Xiang. 2012. "Damage detection by mode shape squares extracted from a passing vehicle." *J. Sound Vib.* 331 (2): 291–307. https://doi.org/10.1016/j.jsv.2011.09.004.

Zheng, F., G. Zhang, and Z. Song. 2001. "Comparison of different implementations of MFCC." *J. Comput. Sci. Technol.* 16 (6): 582–589. https://doi.org/10.1007/BF02943243.

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